

Project - Campus Placement

Campus Placement & Student Performance Analytics

Objective: To analyze student academic performance, skills, internship experience and placement outcomes using Python, NumPy, Pandas and Matplotlib.

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
[2]: df = pd.read_csv("placement.csv")
df
```

```
[2]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
0	STU001	Male	18	BTech	Bengaluru	70.8	2.5	
1	STU002	Male	24	MCA	Noida	77.4	3.6	
2	STU003	Female	18	BSc	Delhi	79.8	8.8	
3	STU004	Female	19	MCA	Delhi	89.3	4.4	
4	STU005	Male	25	BSc	Lucknow	71.7	8.4	
..	...	...	...	...	...	...	...	
995	STU996	Female	20	MCA	Delhi	56.5	4.7	
996	STU997	Female	21	MCA	Delhi	74.8	7.2	
997	STU998	Male	22	MCA	Pune	64.8	4.9	
998	STU999	Male	22	BCA	Mumbai	91.2	6.2	
999	STU1000	Female	18	BTech	Hyderabad	88.7	7.4	

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
0	76.5	83.4	49.9	43.5	
1	80.3	70.8	71.8	60.9	
2	90.1	95.0	85.7	68.8	
3	62.7	81.7	74.8	79.1	
4	73.7	93.1	68.2	56.6	
..	...	...	...	...	
995	63.7	84.0	60.0	83.0	
996	72.4	100.0	67.7	43.3	
997	79.8	94.5	68.5	41.6	
998	75.2	100.0	70.6	83.1	
999	71.3	96.8	77.4	48.9	

Internship	Backlogs	Placement	Package_LPA
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0	Yes	0	Not Placed	0.00
1	Yes	0	Placed	6.27
2	No	0	Placed	5.85
3	No	0	Placed	5.84
4	No	0	Placed	6.06
..	...	...	...	...
995	No	1	Not Placed	0.00
996	Yes	0	Not Placed	0.00
997	No	0	Placed	6.40
998	No	0	Placed	7.42
999	No	0	Not Placed	0.00

[1000 rows x 15 columns]

Data Exploration

```
[3]: # top 5 rows
df.head()
```

[3]:	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
0	STU001	Male	18	BTech	Bengaluru	70.8	2.5	
1	STU002	Male	24	MCA	Noida	77.4	3.6	
2	STU003	Female	18	BSc	Delhi	79.8	8.8	
3	STU004	Female	19	MCA	Delhi	89.3	4.4	
4	STU005	Male	25	BSc	Lucknow	71.7	8.4	

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
0	76.5	83.4	49.9	43.5	
1	80.3	70.8	71.8	60.9	
2	90.1	95.0	85.7	68.8	
3	62.7	81.7	74.8	79.1	
4	73.7	93.1	68.2	56.6	

	Internship	Backlogs	Placement	Package_LPA
0	Yes	0	Not Placed	0.00
1	Yes	0	Placed	6.27
2	No	0	Placed	5.85
3	No	0	Placed	5.84
4	No	0	Placed	6.06

```
[4]: # bottom 5 rows
df.tail()
```

[4]:	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
995	STU996	Female	20	MCA	Delhi	56.5	4.7	
996	STU997	Female	21	MCA	Delhi	74.8	7.2	
997	STU998	Male	22	MCA	Pune	64.8	4.9	
998	STU999	Male	22	BCA	Mumbai	91.2	6.2	

999	STU1000	Female	18	BTech	Hyderabad	88.7	7.4
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	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
995	63.7	84.0	60.0	83.0	
996	72.4	100.0	67.7	43.3	
997	79.8	94.5	68.5	41.6	
998	75.2	100.0	70.6	83.1	
999	71.3	96.8	77.4	48.9	

	Internship	Backlogs	Placement	Package_LPA
995	No	1	Not Placed	0.00
996	Yes	0	Not Placed	0.00
997	No	0	Placed	6.40
998	No	0	Placed	7.42
999	No	0	Not Placed	0.00

```
[5]: # dataframe information
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Student_ID            1000 non-null   str
1   Gender                 1000 non-null   str
2   Age                   1000 non-null   int64
3   Course                 1000 non-null   str
4   City                   1000 non-null   str
5   Attendance             1000 non-null   float64
6   Study_Hours            1000 non-null   float64
7   Assignment_Score       1000 non-null   float64
8   Project_Score          1000 non-null   float64
9   Aptitude_Score         1000 non-null   float64
10  Communication_Score    1000 non-null   float64
11  Internship              1000 non-null   str
12  Backlogs                1000 non-null   int64
13  Placement               1000 non-null   str
14  Package_LPA            1000 non-null   float64
dtypes: float64(7), int64(2), str(6)
memory usage: 117.3 KB
```

```
[6]: df.shape      #(rows, columns)
```

```
[6]: (1000, 15)
```

```
[7]: df.columns
```

```
[7]: Index(['Student_ID', 'Gender', 'Age', 'Course', 'City', 'Attendance',
        'Study_Hours', 'Assignment_Score', 'Project_Score', 'Aptitude_Score',
        'Communication_Score', 'Internship', 'Backlogs', 'Placement',
        'Package_LPA'],
        dtype='str')
```

```
[8]: df.describe()
```

```
[8]:
```

	Age	Attendance	Study_Hours	Assignment_Score	Project_Score	\
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000	
mean	21.478000	78.031800	5.224200	75.065900	83.949600	
std	2.275675	9.514706	2.166797	9.867516	11.244679	
min	18.000000	53.300000	1.500000	45.100000	43.300000	
25%	19.000000	71.300000	3.300000	68.575000	75.700000	
50%	21.000000	78.800000	5.200000	75.050000	84.500000	
75%	23.000000	85.125000	7.100000	81.900000	92.800000	
max	25.000000	98.300000	9.000000	100.000000	100.000000	

	Aptitude_Score	Communication_Score	Backlogs	Package_LPA
count	1000.000000	1000.000000	1000.000000	1000.000000
mean	69.883600	65.872400	0.197000	3.28870
std	10.763203	12.258338	0.659261	3.48665
min	35.000000	40.000000	0.000000	0.00000
25%	62.500000	56.375000	0.000000	0.00000
50%	70.000000	66.200000	0.000000	0.00000
75%	76.825000	75.100000	0.000000	6.78250
max	100.000000	98.900000	4.000000	10.43000

PANDAS OPERATIONS

```
[9]: #attendance values
print(df["Attendance"])
```

```
0    70.8
1    77.4
2    79.8
3    89.3
4    71.7
...
995   56.5
996   74.8
997   64.8
998   91.2
999   88.7
```

```
Name: Attendance, Length: 1000, dtype: float64
```

```
[10]: #multiple columns
print(df[["Student_ID", "Course", "Attendance", "Placement"]])
```

	Student_ID	Course	Attendance	Placement
0	STU001	BTech	70.8	Not Placed
1	STU002	MCA	77.4	Placed
2	STU003	BSc	79.8	Placed
3	STU004	MCA	89.3	Placed
4	STU005	BSc	71.7	Placed
..	...	...	...	...
995	STU996	MCA	56.5	Not Placed
996	STU997	MCA	74.8	Not Placed
997	STU998	MCA	64.8	Placed
998	STU999	BCA	91.2	Placed
999	STU1000	BTech	88.7	Not Placed

[1000 rows x 4 columns]

DATA ANALYSIS

```
[11]: #average attendance
print("Average Attendance =", df["Attendance"].mean())
```

Average Attendance = 78.0318

```
[12]: #average study hours
print("Average Study Hours =", df["Study_Hours"].mean())
```

Average Study Hours = 5.2242000000000001

```
[13]: #average aptitude score
print("Average Aptitude Score =", df["Aptitude_Score"].mean())
```

Average Aptitude Score = 69.8836

```
[14]: #average project score
print("Average Project Score =", df["Project_Score"].mean())
```

Average Project Score = 83.9496

```
[15]: #highest package
print("Highest Package =", df["Package_LPA"].max())
```

Highest Package = 10.43

```
[16]: #lowest attendance
print("Lowest Attendance =", df["Attendance"].min())
```

Lowest Attendance = 53.3

```
[17]: #highest aptitude score
print("Highest Aptitude Score =", df["Aptitude_Score"].max())
```

Highest Aptitude Score = 100.0

TOP STUDENTS

```
[19]: #top 10 students by aptitude
top_students = df.sort_values(by="Aptitude_Score", ascending=False)
top_students.head(10)
```

```
[19]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
466	STU467	Female	22	BCA	Noida	73.8	1.8	
891	STU892	Male	25	BCA	Mumbai	94.9	8.6	
525	STU526	Male	22	BSc	Delhi	70.6	7.6	
696	STU697	Male	23	MCA	Lucknow	78.0	6.9	
653	STU654	Female	23	MCA	Hyderabad	94.1	8.9	
840	STU841	Male	22	BCA	Noida	94.1	8.6	
614	STU615	Male	18	BTech	Hyderabad	82.2	8.8	
374	STU375	Female	25	BTech	Lucknow	76.6	8.7	
963	STU964	Male	21	MCA	Noida	88.5	8.8	
668	STU669	Female	19	BCA	Mumbai	70.0	7.7	

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
466	59.0	60.8	100.0	72.3	
891	74.5	89.4	100.0	46.3	
525	68.6	74.5	100.0	72.4	
696	70.1	84.9	100.0	62.6	
653	92.5	100.0	100.0	48.6	
840	84.4	94.7	99.7	85.2	
614	90.0	93.6	99.0	75.6	
374	85.5	94.1	98.3	79.7	
963	79.1	100.0	97.3	56.6	
668	80.7	100.0	97.3	79.3	

	Internship	Backlogs	Placement	Package_LPA
466	No	3	Not Placed	0.00
891	Yes	0	Placed	9.24
525	No	0	Placed	8.18
696	Yes	0	Placed	7.08
653	No	0	Placed	6.00
840	Yes	0	Placed	7.68
614	No	0	Not Placed	0.00
374	No	0	Placed	6.86
963	No	0	Placed	8.00
668	Yes	0	Placed	7.66

```
[20]: #top 10 project performers
top_project = df.sort_values(by="Project_Score", ascending=False)
top_project.head(10)
```

```
[20]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
	998	STU999	Male	22	BCA	Mumbai	91.2	6.2
	996	STU997	Female	21	MCA	Delhi	74.8	7.2
	975	STU976	Male	18	BCA	Lucknow	72.9	8.7
	963	STU964	Male	21	MCA	Noida	88.5	8.8
	25	STU026	Male	19	BTech	Hyderabad	68.0	8.7
	22	STU023	Male	19	BTech	Lucknow	86.2	8.1
	14	STU015	Male	18	BCA	Pune	92.0	7.9
	946	STU947	Male	20	BCA	Noida	78.2	8.9
	945	STU946	Male	23	MCA	Delhi	89.0	7.5
	936	STU937	Female	24	BCA	Noida	76.7	5.9

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
998	75.2	100.0	70.6	83.1	
996	72.4	100.0	67.7	43.3	
975	85.8	100.0	59.3	56.8	
963	79.1	100.0	97.3	56.6	
25	80.2	100.0	59.8	60.5	
22	76.2	100.0	89.7	88.6	
14	91.6	100.0	79.6	54.0	
946	88.9	100.0	69.7	61.7	
945	88.3	100.0	71.5	59.0	
936	78.4	100.0	54.0	80.8	

	Internship	Backlogs	Placement	Package_LPA
998	No	0	Placed	7.42
996	Yes	0	Not Placed	0.00
975	No	0	Not Placed	0.00
963	No	0	Placed	8.00
25	No	0	Placed	5.76
22	No	0	Not Placed	0.00
14	Yes	0	Placed	7.72
946	No	0	Placed	6.56
945	No	0	Not Placed	0.00
936	No	0	Placed	6.33

```
[21]: #top 10 package holders
top_package = df.sort_values(by="Package_LPA", ascending=False)
top_package.head(10)
```

```
[21]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
	827	STU828	Female	24	BCA	Pune	74.6	8.6
	604	STU605	Female	21	BSc	Delhi	84.1	7.6
	698	STU699	Female	25	MCA	Delhi	80.8	8.5
	891	STU892	Male	25	BCA	Mumbai	94.9	8.6
	706	STU707	Male	21	BTech	Bengaluru	91.9	6.5
	889	STU890	Female	20	BTech	Hyderabad	72.9	7.5

591	STU592	Female	19	BTech	Mumbai	83.9	8.0
488	STU489	Female	21	BSc	Noida	83.8	8.2
829	STU830	Female	20	BSc	Noida	67.8	8.5
845	STU846	Female	21	BSc	Pune	88.1	8.7

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
827	82.1	100.0	66.5	70.8	
604	80.6	85.8	88.5	90.6	
698	86.1	90.2	82.6	87.9	
891	74.5	89.4	100.0	46.3	
706	87.6	89.3	96.4	84.2	
889	72.9	100.0	90.5	67.0	
591	81.7	86.2	71.8	89.5	
488	86.4	91.0	79.2	56.9	
829	73.3	100.0	70.6	84.4	
845	100.0	95.4	85.0	78.2	

	Internship	Backlogs	Placement	Package_LPA
827	Yes	0	Placed	10.43
604	No	0	Placed	9.40
698	Yes	0	Placed	9.25
891	Yes	0	Placed	9.24
706	No	0	Placed	9.18
889	Yes	0	Placed	9.12
591	Yes	0	Placed	9.11
488	Yes	0	Placed	9.11
829	Yes	0	Placed	9.10
845	Yes	0	Placed	9.08

FILTERING

```
[22]: average_attendance = df["Attendance"].mean()
      print(average_attendance)
```

78.0318

```
[23]: #students having attendance above average
      high_attendance = df[df["Attendance"] > average_attendance]
      high_attendance
```

```
[23]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
2	STU003	Female	18	BSc	Delhi	79.8	8.8	
3	STU004	Female	19	MCA	Delhi	89.3	4.4	
5	STU006	Male	23	MCA	Pune	82.7	4.5	
6	STU007	Female	21	MCA	Bengaluru	79.1	2.2	
7	STU008	Female	18	BCA	Lucknow	83.4	5.5	
..	...	...	...	...	...	...	...	
992	STU993	Male	25	BSc	Noida	92.2	8.4	

993	STU994	Female	22	BCA	Bengaluru	79.6	1.8
994	STU995	Female	18	BTech	Hyderabad	85.9	6.5
998	STU999	Male	22	BCA	Mumbai	91.2	6.2
999	STU1000	Female	18	BTech	Hyderabad	88.7	7.4

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
2	90.1	95.0	85.7	68.8	
3	62.7	81.7	74.8	79.1	
5	82.8	80.2	68.2	69.9	
6	71.9	74.0	51.3	55.9	
7	76.7	97.5	79.2	70.7	
..	...	...	...	...	
992	83.8	95.1	73.1	69.3	
993	84.5	84.2	59.7	47.9	
994	89.9	83.7	73.7	57.2	
998	75.2	100.0	70.6	83.1	
999	71.3	96.8	77.4	48.9	

	Internship	Backlogs	Placement	Package_LPA
2	No	0	Placed	5.85
3	No	0	Placed	5.84
5	Yes	0	Not Placed	0.00
6	Yes	0	Not Placed	0.00
7	No	0	Placed	6.34
..	...	...	...	...
992	Yes	0	Placed	7.65
993	No	0	Not Placed	0.00
994	No	0	Placed	5.84
998	No	0	Placed	7.42
999	No	0	Not Placed	0.00

[537 rows x 15 columns]

```
[24]: #students having attendance below average
low_attendance = df[df["Attendance"] < average_attendance]
low_attendance
```

```
[24]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
0	STU001	Male	18	BTech	Bengaluru	70.8	2.5	
1	STU002	Male	24	MCA	Noida	77.4	3.6	
4	STU005	Male	25	BSc	Lucknow	71.7	8.4	
8	STU009	Female	21	MCA	Mumbai	75.2	2.7	
9	STU010	Female	21	BCA	Delhi	70.6	5.8	
..	...	...	...	...	...	...	...	
987	STU988	Female	24	BSc	Mumbai	65.3	6.3	
991	STU992	Female	24	MCA	Hyderabad	77.0	8.6	
995	STU996	Female	20	MCA	Delhi	56.5	4.7	

996	STU997	Female	21	MCA	Delhi	74.8	7.2
997	STU998	Male	22	MCA	Pune	64.8	4.9

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
0	76.5	83.4	49.9		43.5
1	80.3	70.8	71.8		60.9
4	73.7	93.1	68.2		56.6
8	68.1	90.6	57.4		75.6
9	68.8	95.8	77.0		87.1
..	...	...	...	...	
987	83.1	87.6	49.3		48.7
991	86.2	91.9	73.6		60.2
995	63.7	84.0	60.0		83.0
996	72.4	100.0	67.7		43.3
997	79.8	94.5	68.5		41.6

	Internship	Backlogs	Placement	Package_LPA
0	Yes	0	Not Placed	0.00
1	Yes	0	Placed	6.27
4	No	0	Placed	6.06
8	Yes	0	Not Placed	0.00
9	Yes	0	Not Placed	0.00
..	...	...	...	...
987	Yes	1	Placed	6.41
991	No	1	Not Placed	0.00
995	No	1	Not Placed	0.00
996	Yes	0	Not Placed	0.00
997	No	0	Placed	6.40

[463 rows x 15 columns]

```
[25]: average_apititude = df["Aptitude_Score"].mean()
print(average_apititude)
```

69.8836

```
[26]: #students above average aptitude
high_apititude = df[df["Aptitude_Score"] > average_apititude]
high_apititude
```

```
[26]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
1	STU002	Male	24	MCA	Noida	77.4	3.6	
2	STU003	Female	18	BSc	Delhi	79.8	8.8	
3	STU004	Female	19	MCA	Delhi	89.3	4.4	
7	STU008	Female	18	BCA	Lucknow	83.4	5.5	
9	STU010	Female	21	BCA	Delhi	70.6	5.8	
..	...	...	...	...	...	...		
991	STU992	Female	24	MCA	Hyderabad	77.0	8.6	

992	STU993	Male	25	BSc	Noida	92.2	8.4
994	STU995	Female	18	BTech	Hyderabad	85.9	6.5
998	STU999	Male	22	BCA	Mumbai	91.2	6.2
999	STU1000	Female	18	BTech	Hyderabad	88.7	7.4

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
1	80.3	70.8	71.8	60.9	
2	90.1	95.0	85.7	68.8	
3	62.7	81.7	74.8	79.1	
7	76.7	97.5	79.2	70.7	
9	68.8	95.8	77.0	87.1	
..	...	...	...	...	
991	86.2	91.9	73.6	60.2	
992	83.8	95.1	73.1	69.3	
994	89.9	83.7	73.7	57.2	
998	75.2	100.0	70.6	83.1	
999	71.3	96.8	77.4	48.9	

	Internship	Backlogs	Placement	Package_LPA
1	Yes	0	Placed	6.27
2	No	0	Placed	5.85
3	No	0	Placed	5.84
7	No	0	Placed	6.34
9	Yes	0	Not Placed	0.00
..	...	...	...	...
991	No	1	Not Placed	0.00
992	Yes	0	Placed	7.65
994	No	0	Placed	5.84
998	No	0	Placed	7.42
999	No	0	Not Placed	0.00

[508 rows x 15 columns]

```
[27]: #students who completed internship
internship_students = df[df["Internship"] == "Yes"]
internship_students
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
0	STU001	Male	18	BTech	Bengaluru	70.8	2.5	
1	STU002	Male	24	MCA	Noida	77.4	3.6	
5	STU006	Male	23	MCA	Pune	82.7	4.5	
6	STU007	Female	21	MCA	Bengaluru	79.1	2.2	
8	STU009	Female	21	MCA	Mumbai	75.2	2.7	
..	...	...	...	...	...	...	...	
980	STU981	Male	23	BCA	Pune	77.0	7.6	
981	STU982	Female	21	BTech	Pune	70.7	7.5	
987	STU988	Female	24	BSc	Mumbai	65.3	6.3	

992	STU993	Male	25	BSc	Noida	92.2	8.4
996	STU997	Female	21	MCA	Delhi	74.8	7.2

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
0	76.5	83.4	49.9		43.5
1	80.3	70.8	71.8		60.9
5	82.8	80.2	68.2		69.9
6	71.9	74.0	51.3		55.9
8	68.1	90.6	57.4		75.6
..	...	...	...	...	
980	86.1	100.0	97.2		52.3
981	75.7	87.2	84.3		72.7
987	83.1	87.6	49.3		48.7
992	83.8	95.1	73.1		69.3
996	72.4	100.0	67.7		43.3

	Internship	Backlogs	Placement	Package_LPA
0	Yes	0	Not Placed	0.00
1	Yes	0	Placed	6.27
5	Yes	0	Not Placed	0.00
6	Yes	0	Not Placed	0.00
8	Yes	0	Not Placed	0.00
..	...	...	...	...
980	Yes	0	Placed	8.98
981	Yes	0	Placed	7.28
987	Yes	1	Placed	6.41
992	Yes	0	Placed	7.65
996	Yes	0	Not Placed	0.00

[407 rows x 15 columns]

```
[28]: #placed students
placed_students = df[df["Placement"] == "Placed"]
placed_students
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
1	STU002	Male	24	MCA	Noida	77.4	3.6	
2	STU003	Female	18	BSc	Delhi	79.8	8.8	
3	STU004	Female	19	MCA	Delhi	89.3	4.4	
4	STU005	Male	25	BSc	Lucknow	71.7	8.4	
7	STU008	Female	18	BCA	Lucknow	83.4	5.5	
..	...	...	...	...	...	...		
990	STU991	Male	24	BCA	Noida	85.9	6.2	
992	STU993	Male	25	BSc	Noida	92.2	8.4	
994	STU995	Female	18	BTech	Hyderabad	85.9	6.5	
997	STU998	Male	22	MCA	Pune	64.8	4.9	
998	STU999	Male	22	BCA	Mumbai	91.2	6.2	

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
1	80.3	70.8	71.8	60.9	
2	90.1	95.0	85.7	68.8	
3	62.7	81.7	74.8	79.1	
4	73.7	93.1	68.2	56.6	
7	76.7	97.5	79.2	70.7	
..	...	...	...	...	
990	76.6	85.1	55.4	66.1	
992	83.8	95.1	73.1	69.3	
994	89.9	83.7	73.7	57.2	
997	79.8	94.5	68.5	41.6	
998	75.2	100.0	70.6	83.1	

	Internship	Backlogs	Placement	Package_LPA
1	Yes	0	Placed	6.27
2	No	0	Placed	5.85
3	No	0	Placed	5.84
4	No	0	Placed	6.06
7	No	0	Placed	6.34
..	...	...	...	...
990	No	0	Placed	5.96
992	Yes	0	Placed	7.65
994	No	0	Placed	5.84
997	No	0	Placed	6.40
998	No	0	Placed	7.42

[480 rows x 15 columns]

```
[29]: #not placed students
not_placed_students = df[df["Placement"] == "Not Placed"]
not_placed_students
```

[29]:	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
0	STU001	Male	18	BTech	Bengaluru	70.8	2.5	
5	STU006	Male	23	MCA	Pune	82.7	4.5	
6	STU007	Female	21	MCA	Bengaluru	79.1	2.2	
8	STU009	Female	21	MCA	Mumbai	75.2	2.7	
9	STU010	Female	21	BCA	Delhi	70.6	5.8	
..	...	...	...	...	...	...	...	
991	STU992	Female	24	MCA	Hyderabad	77.0	8.6	
993	STU994	Female	22	BCA	Bengaluru	79.6	1.8	
995	STU996	Female	20	MCA	Delhi	56.5	4.7	
996	STU997	Female	21	MCA	Delhi	74.8	7.2	
999	STU1000	Female	18	BTech	Hyderabad	88.7	7.4	

Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
------------------	---------------	----------------	---------------------	---

0	76.5	83.4	49.9	43.5
5	82.8	80.2	68.2	69.9
6	71.9	74.0	51.3	55.9
8	68.1	90.6	57.4	75.6
9	68.8	95.8	77.0	87.1
..	...	...	...	...
991	86.2	91.9	73.6	60.2
993	84.5	84.2	59.7	47.9
995	63.7	84.0	60.0	83.0
996	72.4	100.0	67.7	43.3
999	71.3	96.8	77.4	48.9

	Internship	Backlogs	Placement	Package_LPA
0	Yes	0	Not Placed	0.0
5	Yes	0	Not Placed	0.0
6	Yes	0	Not Placed	0.0
8	Yes	0	Not Placed	0.0
9	Yes	0	Not Placed	0.0
..	...	...	...	...
991	No	1	Not Placed	0.0
993	No	0	Not Placed	0.0
995	No	1	Not Placed	0.0
996	Yes	0	Not Placed	0.0
999	No	0	Not Placed	0.0

[520 rows x 15 columns]

```
[30]: #students having backlogs
backlog_students = df[df["Backlogs"] > 0]
backlog_students
```

```
[30]: Student_ID  Gender  Age  Course  City  Attendance  Study_Hours  \
13      STU014   Male   24   BTech   Lucknow      86.9         2.3
17      STU018  Female   22    MCA   Lucknow      83.1         3.9
35      STU036  Female   24   BSc    Lucknow      58.5         4.0
47      STU048  Female   19   BCA    Mumbai      76.0         2.1
74      STU075   Male   19   BCA    Noida       65.8         4.2
..      ...      ...    ...    ...    ...      ...
967     STU968   Male   21   BTech    Pune      80.7         3.6
984     STU985   Male   22    MCA   Hyderabad   61.7         6.0
987     STU988  Female   24   BSc    Mumbai      65.3         6.3
991     STU992  Female   24    MCA   Hyderabad   77.0         8.6
995     STU996  Female   20    MCA    Delhi      56.5         4.7

Assignment_Score  Project_Score  Aptitude_Score  Communication_Score  \
13              74.7           83.5           69.3             64.6
17              73.4           83.6           61.0             76.4
```

35	71.9	75.7	54.3	52.2
47	69.2	69.7	60.5	49.7
74	71.1	69.2	73.1	50.0
..	...	...	...	...
967	74.1	78.7	72.9	49.0
984	78.2	87.6	82.4	77.4
987	83.1	87.6	49.3	48.7
991	86.2	91.9	73.6	60.2
995	63.7	84.0	60.0	83.0

	Internship	Backlogs	Placement	Package_LPA
13	No	3	Not Placed	0.00
17	No	2	Not Placed	0.00
35	No	1	Not Placed	0.00
47	No	2	Not Placed	0.00
74	No	1	Not Placed	0.00
..	...	...	...	...
967	No	1	Placed	6.98
984	No	3	Not Placed	0.00
987	Yes	1	Placed	6.41
991	No	1	Not Placed	0.00
995	No	1	Not Placed	0.00

[100 rows x 15 columns]

MULTIPLE FILTERING

```
[31]: #students having low attendance and low aptitude
students_need_attention = df[
    (df["Attendance"] < 70) &
    (df["Aptitude_Score"] < 60)
]
students_need_attention
```

```
[31]:
```

	Student_ID	Gender	Age	Course	City	Attendance	Study_Hours	\
25	STU026	Male	19	BTech	Hyderabad	68.0	8.7	
34	STU035	Male	24	BSc	Delhi	54.9	4.1	
35	STU036	Female	24	BSc	Lucknow	58.5	4.0	
49	STU050	Male	20	MCA	Hyderabad	67.8	2.1	
78	STU079	Male	19	BCA	Mumbai	63.6	2.2	
110	STU111	Male	21	BSc	Bengaluru	58.2	5.7	
111	STU112	Female	24	BCA	Delhi	65.4	3.9	
122	STU123	Male	18	BSc	Delhi	67.7	6.2	
140	STU141	Male	21	BCA	Bengaluru	64.1	1.8	
142	STU143	Male	24	BSc	Delhi	60.2	4.1	
159	STU160	Male	24	MCA	Delhi	66.1	3.8	
165	STU166	Female	23	BSc	Hyderabad	64.4	2.2	
201	STU202	Male	19	BTech	Pune	65.3	8.6	

205	STU206	Female	23	MCA	Delhi	67.9	5.9
212	STU213	Male	25	MCA	Bengaluru	67.3	2.1
240	STU241	Female	23	MCA	Noida	54.0	6.5
253	STU254	Female	23	MCA	Noida	69.0	6.2
255	STU256	Male	22	BTech	Delhi	66.5	2.0
259	STU260	Female	25	BTech	Delhi	62.2	6.8
301	STU302	Female	24	MCA	Bengaluru	68.3	1.7
303	STU304	Male	20	BCA	Bengaluru	59.1	4.2
323	STU324	Female	20	BCA	Hyderabad	59.4	4.2
388	STU389	Female	21	MCA	Lucknow	59.2	1.7
400	STU401	Male	20	MCA	Noida	63.2	7.9
409	STU410	Female	18	BTech	Hyderabad	68.5	1.7
423	STU424	Female	22	BTech	Delhi	63.7	5.9
446	STU447	Male	20	BTech	Bengaluru	66.0	3.4
495	STU496	Female	22	BSc	Bengaluru	58.3	2.6
506	STU507	Male	22	BSc	Noida	69.4	5.3
512	STU513	Male	22	BSc	Bengaluru	53.7	4.7
519	STU520	Female	20	MCA	Pune	66.1	1.8
612	STU613	Male	20	BCA	Hyderabad	67.2	3.1
616	STU617	Female	21	BSc	Lucknow	69.8	3.5
619	STU620	Male	18	BCA	Mumbai	65.8	7.3
628	STU629	Male	21	MCA	Lucknow	61.0	2.5
650	STU651	Male	19	BCA	Hyderabad	63.4	1.6
688	STU689	Male	24	BSc	Bengaluru	63.8	5.4
736	STU737	Female	19	BTech	Delhi	68.6	5.7
757	STU758	Female	21	MCA	Mumbai	54.0	3.2
771	STU772	Male	23	BTech	Hyderabad	67.3	5.6
778	STU779	Female	20	BSc	Mumbai	66.7	3.8
783	STU784	Male	24	MCA	Mumbai	67.9	5.5
790	STU791	Female	25	BCA	Pune	67.0	5.9
808	STU809	Female	19	BSc	Delhi	63.0	3.0
853	STU854	Male	24	BTech	Lucknow	66.5	4.1
865	STU866	Male	20	BSc	Bengaluru	58.6	1.8
866	STU867	Male	19	BSc	Lucknow	68.7	2.7
884	STU885	Male	21	MCA	Hyderabad	69.7	2.9
901	STU902	Male	19	BCA	Bengaluru	61.5	4.1
927	STU928	Female	21	BTech	Hyderabad	57.9	3.5
932	STU933	Male	23	BSc	Lucknow	65.0	7.0
940	STU941	Female	25	MCA	Delhi	54.9	7.3
951	STU952	Male	23	BTech	Hyderabad	68.6	4.0
956	STU957	Male	18	BCA	Mumbai	55.8	4.0
987	STU988	Female	24	BSc	Mumbai	65.3	6.3

	Assignment_Score	Project_Score	Aptitude_Score	Communication_Score	\
25	80.2	100.0	59.8	60.5	
34	65.1	77.5	43.5	68.2	
35	71.9	75.7	54.3	52.2	

49	72.7	72.9	54.5	49.9
78	61.7	54.7	48.4	61.4
110	59.8	91.5	53.8	44.9
111	75.6	84.1	59.4	78.6
122	90.2	82.1	56.5	49.4
140	71.4	61.1	51.2	46.3
142	61.5	69.7	48.8	57.6
159	64.7	76.1	54.9	64.9
165	65.4	83.6	46.7	53.9
201	96.6	100.0	59.3	66.9
205	72.2	90.2	54.5	92.7
212	67.6	65.5	51.0	63.2
240	68.9	99.5	57.2	75.2
253	72.8	94.3	56.6	55.3
255	64.2	70.6	54.4	59.1
259	54.0	79.5	52.4	54.0
301	58.2	64.7	46.1	82.0
303	80.0	74.6	44.2	80.6
323	71.2	85.5	57.8	75.9
388	51.4	69.8	58.9	66.3
400	50.5	87.1	56.2	47.5
409	68.1	55.4	51.8	69.5
423	72.6	84.9	58.7	66.2
446	60.0	76.8	49.8	62.0
495	68.2	74.7	50.5	80.8
506	80.2	92.0	48.4	65.2
512	63.5	73.3	51.2	68.9
519	70.0	78.4	46.0	61.7
612	77.8	72.7	43.9	62.1
616	54.2	58.6	55.6	79.2
619	74.2	100.0	57.1	40.0
628	74.7	71.9	59.3	79.9
650	64.6	66.7	59.6	54.3
688	88.3	86.5	56.2	64.9
736	77.0	92.8	56.8	79.6
757	65.8	92.2	56.9	64.1
771	73.7	85.5	57.6	57.6
778	76.1	75.8	54.4	63.4
783	75.7	88.9	47.1	57.7
790	65.9	76.5	54.2	82.3
808	59.2	69.8	49.4	71.1
853	65.9	80.8	50.7	79.7
865	47.3	43.3	53.3	80.1
866	60.3	65.8	48.7	77.0
884	72.2	90.7	55.9	68.3
901	59.1	73.8	50.4	84.1
927	62.2	78.1	57.2	75.8

932	78.2	82.2	59.1	65.0
940	79.4	99.4	49.1	81.3
951	85.1	80.9	58.1	63.7
956	72.0	74.6	57.4	61.2
987	83.1	87.6	49.3	48.7

	Internship	Backlogs	Placement	Package_LPA
25	No	0	Placed	5.76
34	No	0	Not Placed	0.00
35	No	1	Not Placed	0.00
49	No	0	Not Placed	0.00
78	Yes	1	Not Placed	0.00
110	Yes	0	Not Placed	0.00
111	No	0	Not Placed	0.00
122	No	1	Not Placed	0.00
140	Yes	3	Not Placed	0.00
142	Yes	2	Not Placed	0.00
159	Yes	0	Placed	5.78
165	No	0	Not Placed	0.00
201	Yes	0	Placed	7.69
205	No	0	Not Placed	0.00
212	Yes	0	Not Placed	0.00
240	No	0	Placed	6.65
253	No	0	Not Placed	0.00
255	No	0	Not Placed	0.00
259	No	0	Not Placed	0.00
301	Yes	1	Not Placed	0.00
303	No	3	Not Placed	0.00
323	Yes	0	Not Placed	0.00
388	No	0	Not Placed	0.00
400	No	0	Not Placed	0.00
409	No	0	Not Placed	0.00
423	Yes	0	Placed	7.15
446	No	3	Not Placed	0.00
495	Yes	0	Placed	7.18
506	No	0	Not Placed	0.00
512	No	2	Not Placed	0.00
519	No	0	Not Placed	0.00
612	No	0	Not Placed	0.00
616	No	0	Not Placed	0.00
619	No	0	Not Placed	0.00
628	No	0	Not Placed	0.00
650	No	0	Not Placed	0.00
688	No	0	Not Placed	0.00
736	No	0	Placed	5.84
757	Yes	4	Not Placed	0.00
771	No	0	Not Placed	0.00

778	No	0	Placed	5.57
783	No	0	Not Placed	0.00
790	Yes	0	Not Placed	0.00
808	No	3	Not Placed	0.00
853	No	0	Not Placed	0.00
865	Yes	0	Placed	5.35
866	No	0	Not Placed	0.00
884	No	0	Not Placed	0.00
901	No	2	Not Placed	0.00
927	No	0	Not Placed	0.00
932	No	0	Not Placed	0.00
940	Yes	0	Not Placed	0.00
951	No	0	Not Placed	0.00
956	No	0	Not Placed	0.00
987	Yes	1	Placed	6.41

COURSE WISE ANALYSIS

```
[32]: #average aptitude by course
print(df.groupby("Course")["Aptitude_Score"].mean())
```

```
Course
BCA      70.843443
BSc      69.217717
BTech    70.110811
MCA      69.471071
Name: Aptitude_Score, dtype: float64
```

```
[33]: #average attendance by course
print(df.groupby("Course")["Attendance"].mean())
```

```
Course
BCA      77.790574
BSc      78.556693
BTech    77.880180
MCA      77.886071
Name: Attendance, dtype: float64
```

```
[34]: #highest aptitude score by course
print(df.groupby("Course")["Aptitude_Score"].max())
```

```
Course
BCA      100.0
BSc      100.0
BTech     99.0
MCA      100.0
Name: Aptitude_Score, dtype: float64
```

GENDER WISE ANALYSIS

```
[35]: #count of each gender
print(df["Gender"].value_counts())
```

```
Gender
Female    505
Male      495
Name: count, dtype: int64
```

```
[36]: #gender wise average aptitude
print(df.groupby("Gender")["Aptitude_Score"].mean())
```

```
Gender
Female    69.829505
Male      69.938788
Name: Aptitude_Score, dtype: float64
```

```
[37]: #gender wise average attendance
print(df.groupby("Gender")["Attendance"].mean())
```

```
Gender
Female    78.134455
Male      77.927071
Name: Attendance, dtype: float64
```

```
[38]: #placement count
placement_count = df["Placement"].value_counts()
placement_count
```

```
[38]: Placement
Not Placed    520
Placed        480
Name: count, dtype: int64
```

```
[39]: #placement percentage
total_students = len(df)
placed_count = df[df["Placement"] == "Placed"].shape[0]
placement_percentage = (placed_count / total_students) * 100
print("Total Students =", total_students)
print("Placed Students =", placed_count)
print("Placement Percentage =", placement_percentage)
```

```
Total Students = 1000
Placed Students = 480
Placement Percentage = 48.0
```

```
[40]: #course wise placement count
course_placement = df[df["Placement"] == "Placed"]["Course"].value_counts()
course_placement
```

```
[40]: Course
      BSc      136
      MCA      121
      BTech    118
      BCA      105
      Name: count, dtype: int64
```

```
[41]: #placement status of internship students
internship_placement = df[df["Internship"] == "Yes"]["Placement"].value_counts()
internship_placement
```

```
[41]: Placement
      Placed      270
      Not Placed  137
      Name: count, dtype: int64
```

```
[42]: #placement status of students without internship
no_internship_placement = df[df["Internship"] == "No"]["Placement"].
    ↪value_counts()
no_internship_placement
```

```
[42]: Placement
      Not Placed  383
      Placed      210
      Name: count, dtype: int64
```

```
[43]: #average package of placed students
print(
    df[df["Placement"] == "Placed"]["Package_LPA"].mean()
)
```

6.8514583333333325

```
[44]: #highest package holder
highest_package = df.sort_values(by="Package_LPA", ascending=False)
highest_package.head(1)
```

```
[44]:   Student_ID  Gender  Age  Course  City  Attendance  Study_Hours  \
827      STU828  Female   24    BCA  Pune         74.6           8.6

      Assignment_Score  Project_Score  Aptitude_Score  Communication_Score  \
827                82.1           100.0           66.5                70.8

      Internship  Backlogs  Placement  Package_LPA
827          Yes         0     Placed         10.43
```

NUMPY ANALYSIS

```
[45]: aptitude = np.array(df["Aptitude_Score"])
```

```
[46]: print("Minimum Aptitude Score =", np.min(aptitude))
      print("Maximum Aptitude Score =", np.max(aptitude))
      print("Total Aptitude Score =", np.sum(aptitude))
      print("Average Aptitude Score =", np.mean(aptitude))
```

```
Minimum Aptitude Score = 35.0
Maximum Aptitude Score = 100.0
Total Aptitude Score = 69883.6
Average Aptitude Score = 69.8836
```

```
[47]: high_scores = aptitude[aptitude > 80]
      print(high_scores)
```

```
[ 85.7  81.   89.7  86.2  83.4  87.   82.8  86.9  81.4  94.2  86.   90.3
  93.2  82.1  83.3  86.4  83.8  84.   89.2  88.2  88.9  81.8  87.4  86.2
  84.4  84.3  81.2  91.5  84.6  81.7  83.1  82.4  85.7  83.5  93.7  96.2
  83.4  81.4  86.5  86.9  85.6  83.3  88.9  85.7  83.9  95.3  84.   92.5
  81.5  87.8  81.7  83.4  80.7  82.9  92.9  90.6  82.4  98.3  87.5  85.7
  83.8  94.1  86.6  82.   87.9  82.3  82.3  80.5  82.   82.5 100.   85.3
  82.1  86.1  84.3  80.9  81.1  86.2  81.6  81.3  80.7 100.   80.3  85.2
  86.9  84.   81.5  82.6  87.6  83.1  81.1  92.4  81.9  81.1  89.4  86.6
  82.9  83.   81.5  88.5  82.3  85.6  83.3  88.5  82.4  99.   84.1  80.3
  83.5  82.1  86.4  89.5  85.2  83.8  84.1 100.   84.3  80.5  97.3  83.1
  83.2 100.   82.5  82.6  90.3  96.4  83.5  81.5  93.8  80.4  87.4  90.9
  83.   81.4  81.6  83.5  80.2  82.3  99.7  83.2  85.   87.2  86.3  80.2
  90.5 100.   84.1  84.1  82.2  84.4  88.   82.3  80.4  85.6  82.   85.
  80.4  85.7  97.3  96.2  84.5  85.5  90.7  96.2  97.2  84.3  97.1  82.4]
```

```
[48]: print("Number of students scoring above 80 =", len(high_scores))
```

```
Number of students scoring above 80 = 168
```

```
[49]: #placement ready
      def placement_readiness(score):
          if score >= 80:
              return "High"
          elif score >= 60:
              return "Moderate"
          else:
              return "Needs Improvement"
```

```
[50]: df["Overall_Score"] = (
      df["Assignment_Score"] +
      df["Project_Score"] +
      df["Aptitude_Score"] +
      df["Communication_Score"]
      ) / 4
```

```
[51]: df["Readiness"] = df["Overall_Score"].apply(placement_readiness)
df[["Student_ID", "Overall_Score", "Readiness"]].head(10)
```

```
[51]:
```

	Student_ID	Overall_Score	Readiness
0	STU001	63.325	Moderate
1	STU002	70.950	Moderate
2	STU003	84.900	High
3	STU004	74.575	Moderate
4	STU005	72.900	Moderate
5	STU006	75.275	Moderate
6	STU007	63.275	Moderate
7	STU008	81.025	High
8	STU009	72.925	Moderate
9	STU010	82.175	High

```
[52]: readiness_count = df["Readiness"].value_counts()
readiness_count
```

```
[52]:
```

Readiness	
Moderate	809
High	175
Needs Improvement	16

Name: count, dtype: int64

GRAPHS

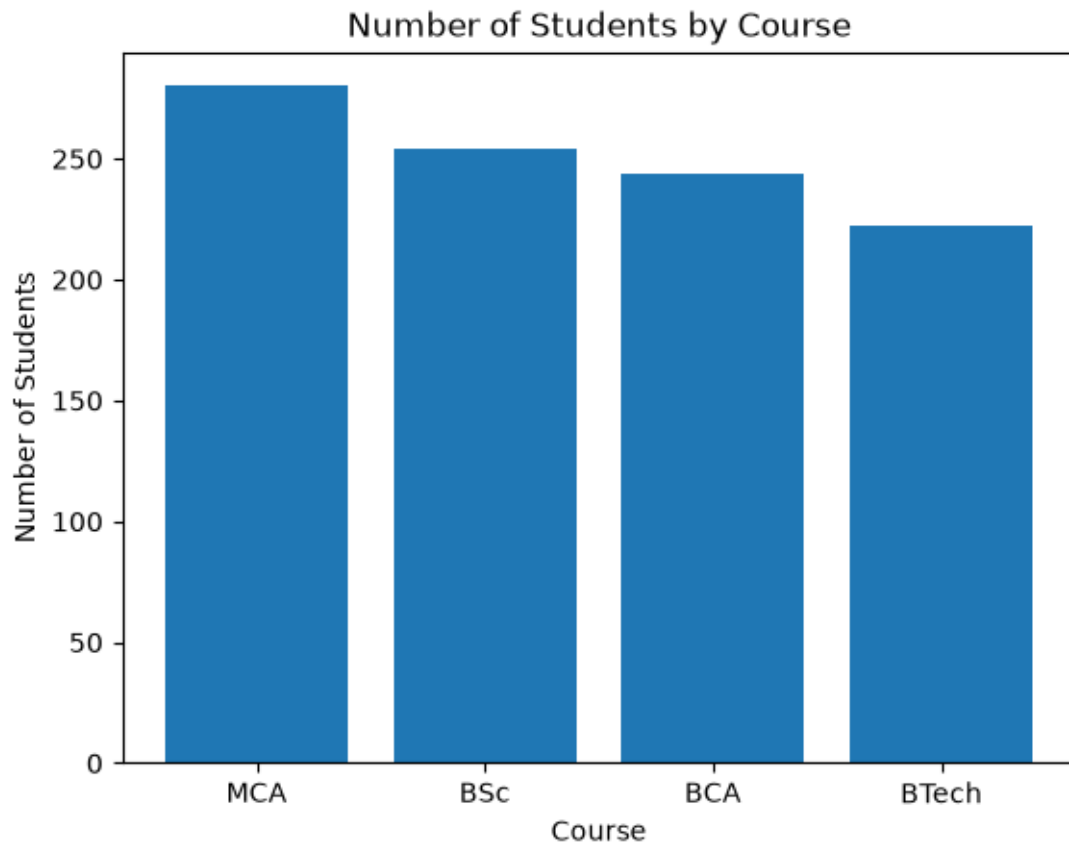
```
[53]: course_count = df["Course"].value_counts()
course_count
```

```
[53]:
```

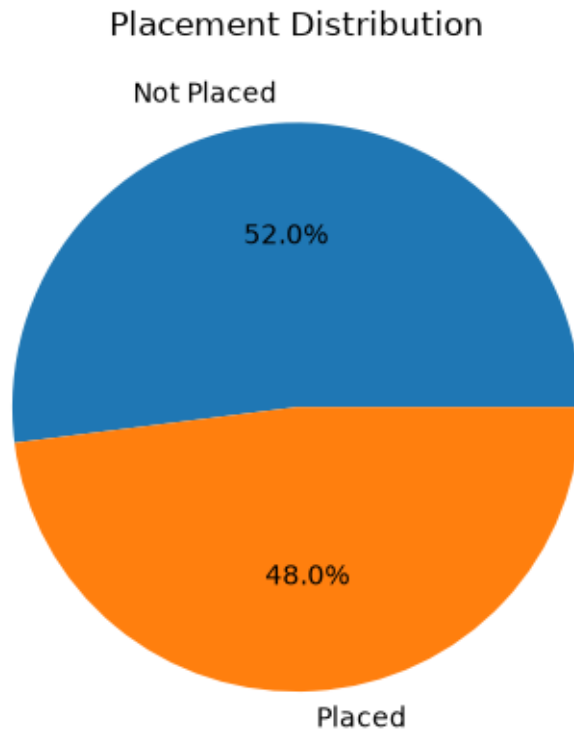
Course	
MCA	280
BSc	254
BCA	244
BTech	222

Name: count, dtype: int64

```
[54]: plt.bar(course_count.index, course_count.values)
plt.title("Number of Students by Course")
plt.xlabel("Course")
plt.ylabel("Number of Students")
plt.show()
```



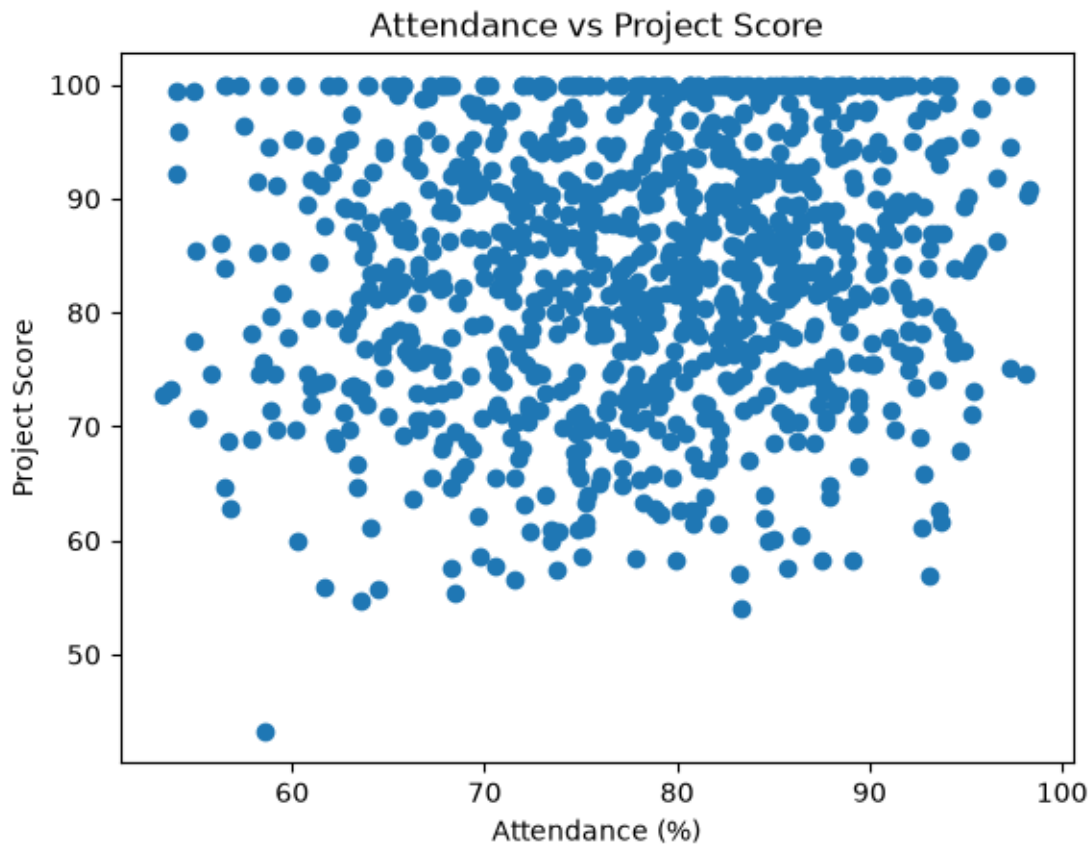
```
[55]: plt.pie(  
    placement_count.values,  
    labels=placement_count.index,  
    autopct="%1.1f%%"  
)  
  
plt.title("Placement Distribution")  
plt.show()
```

```
[56]: course_aptitude = df.groupby("Course")["Aptitude_Score"].mean()  
course_aptitude
```

```
[56]: Course  
BCA      70.843443  
BSc      69.217717  
BTech    70.110811  
MCA      69.471071  
Name: Aptitude_Score, dtype: float64
```

```
[57]: plt.scatter(df["Attendance"], df["Project_Score"])  
  
plt.title("Attendance vs Project Score")  
plt.xlabel("Attendance (%)")  
plt.ylabel("Project Score")  
  
plt.show()
```

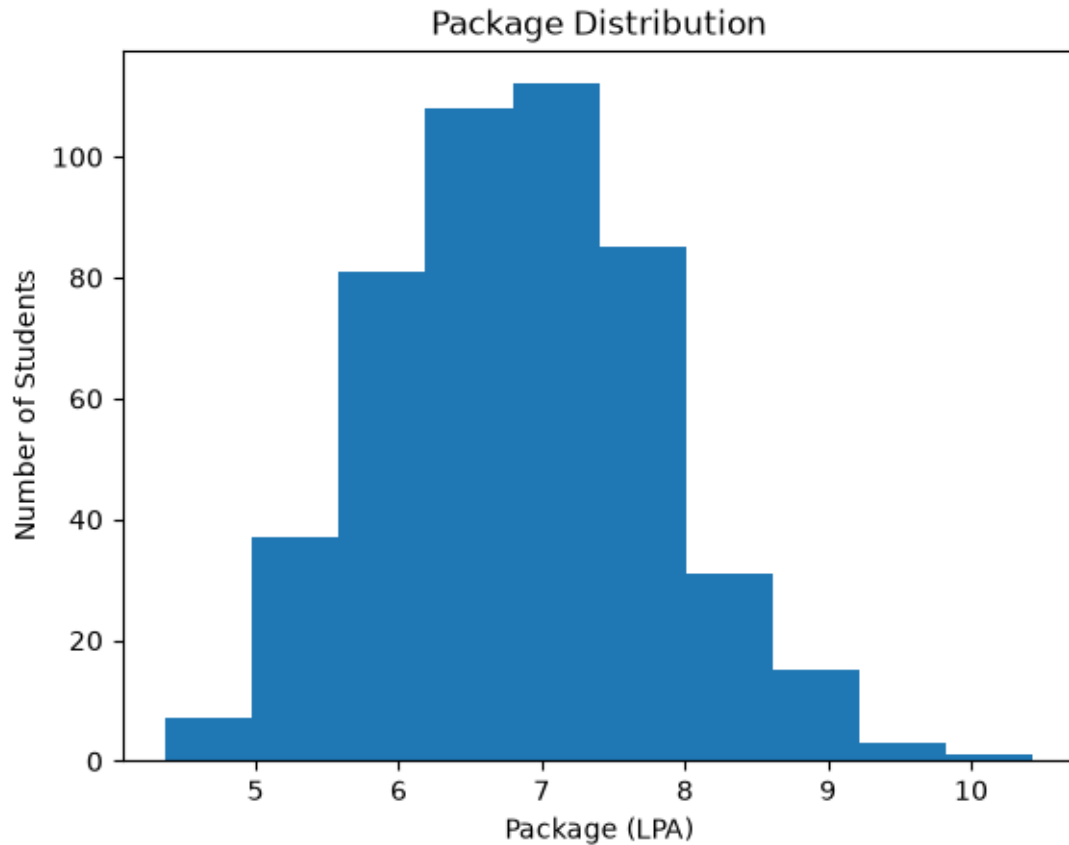


```
[58]: placed_data = df[df["Placement"] == "Placed"]

plt.hist(placed_data["Package_LPA"])

plt.title("Package Distribution")
plt.xlabel("Package (LPA)")
plt.ylabel("Number of Students")

plt.show()
```



```

readiness_count = df["Readiness"].value_counts()
plt.bar(readiness_count.index, readiness_count.values)

plt.title("Placement Readiness of Students") plt.xlabel("Readiness Level") plt.ylabel("Number of Students")

plt.show()

```

Course-wise Attendance vs Project Score

```

[61]: course_attendance = df.groupby("Course")["Attendance"].mean()
      course_project = df.groupby("Course")["Project_Score"].mean()

```

```

[62]: plt.plot(
      course_attendance.index,
      course_attendance.values,
      marker="o",
      label="Attendance"
    )

plt.plot(

```

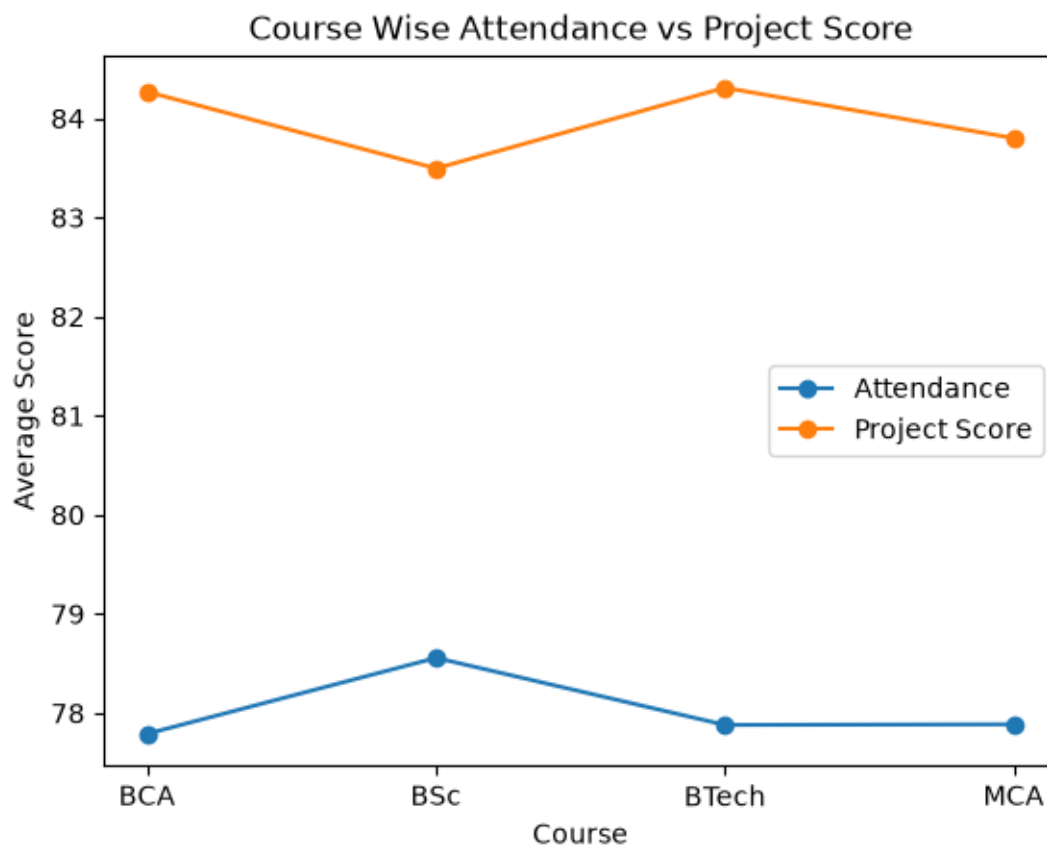
```

course_project.index,
course_project.values,
marker="o",
label="Project Score"
)

plt.title("Course Wise Attendance vs Project Score")
plt.xlabel("Course")
plt.ylabel("Average Score")
plt.legend()

plt.show()

```



Internship Students Performance Comparison

```

[63]: internship_apititude = df.groupby("Internship")["Aptitude_Score"].mean()
      internship_project = df.groupby("Internship")["Project_Score"].mean()

```

```

[64]: plt.bar(
      internship_apititude.index,

```

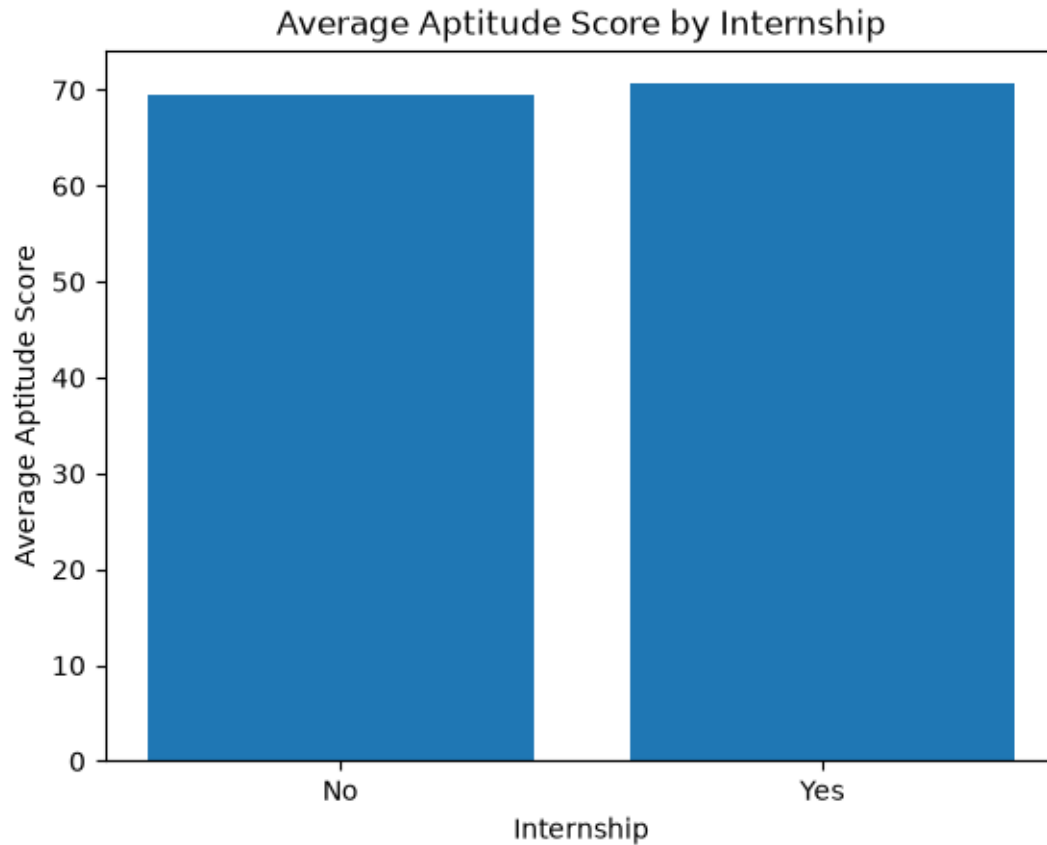
```

    internship_apptitude.values
)

plt.title("Average Aptitude Score by Internship")
plt.xlabel("Internship")
plt.ylabel("Average Aptitude Score")

plt.show()

```



Readiness Dashboard

```
[66]: df["Readiness"]
```

```

[66]: 0    Moderate
      1    Moderate
      2      High
      3    Moderate
      4    Moderate
      ...
     995  Moderate

```

```
996    Moderate
997    Moderate
998        High
999    Moderate
Name: Readiness, Length: 1000, dtype: str
```

```
[67]: # READINESS WISE PLACEMENT

readiness_count = df["Readiness"].value_counts()

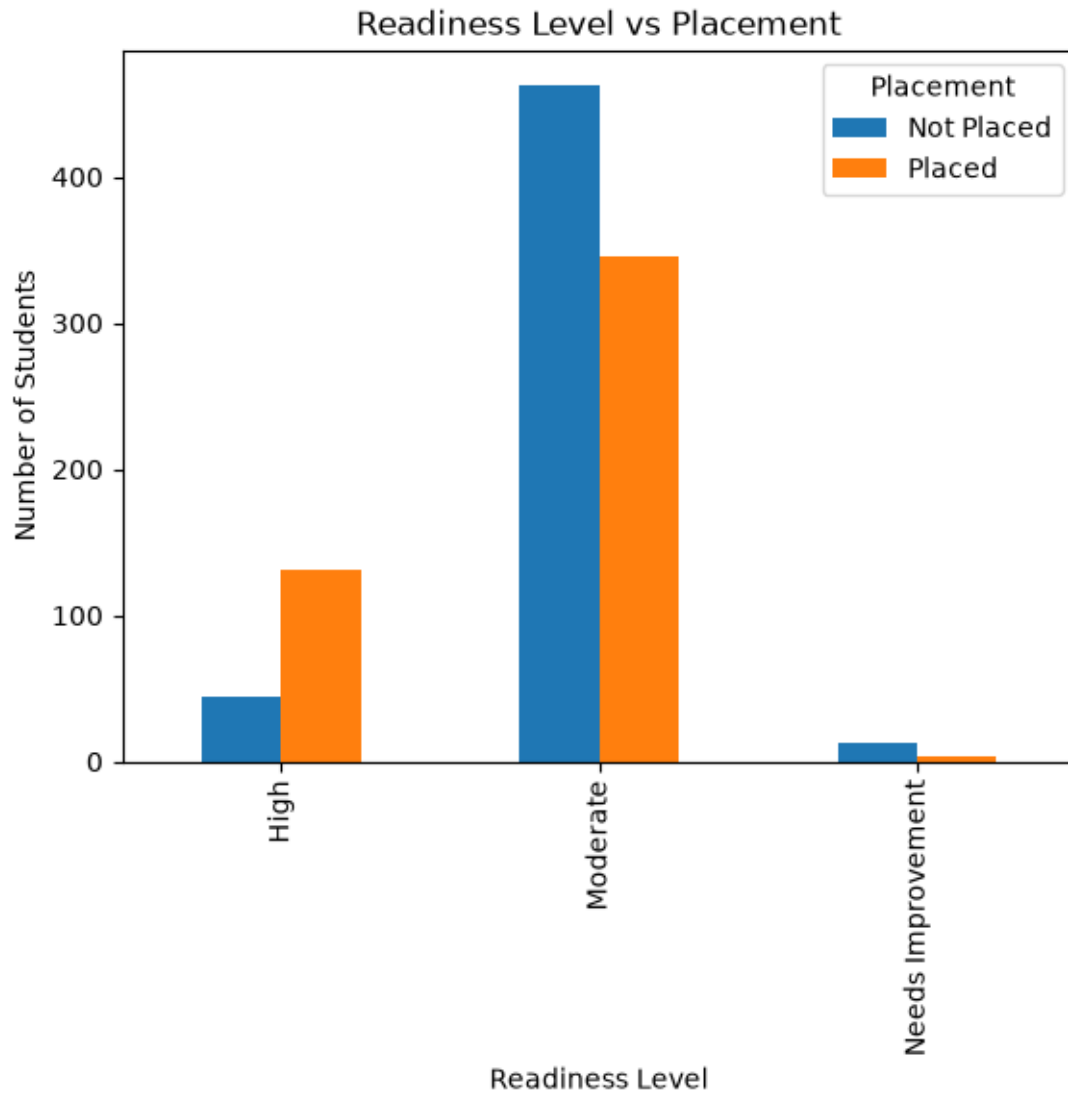
readiness_placement = pd.crosstab(
    df["Readiness"],
    df["Placement"]
)

readiness_placement
```

```
[67]: Placement          Not Placed   Placed
Readiness
High                44         131
Moderate            463         346
Needs Improvement   13           3
```

```
[68]: readiness_placement.plot(
    kind="bar"
)

plt.title("Readiness Level vs Placement")
plt.xlabel("Readiness Level")
plt.ylabel("Number of Students")
plt.show()
```



```
[60]: print("Campus Placement & Student Performance Analytics")
print("-----")
print("Total Students =", total_students)
print("Placed Students =", placed_count)
print("Placement Percentage =", placement_percentage)
print("Average Attendance =", df["Attendance"].mean())
print("Average Aptitude Score =", df["Aptitude_Score"].mean())
print("Highest Package =", df["Package_LPA"].max())
```

Campus Placement & Student Performance Analytics

Total Students = 1000
Placed Students = 480

Placement Percentage = 48.0
Average Attendance = 78.0318
Average Aptitude Score = 69.8836
Highest Package = 10.43

OVERALL ANALYSIS

0.1 Overall Analysis

The dataset contains 1000 student records covering academic performance, attendance, skills, internship experience and placement outcomes.

Out of 1000 students, 480 students were placed, giving an overall placement rate of 48%. The average attendance is around 78%, while the average aptitude score is around 70%.

The analysis also shows a noticeable difference in placement outcomes between students with internship experience and those without internships. Course-wise placement counts also differ, so placement rate provides a better basis for comparison than raw placement count alone.

The project further uses a simple rule-based readiness indicator to group students based on their overall academic and skill scores. This is intended as an analytical indicator rather than a machine-learning prediction model.